



MATH-101 – Discrete Mathematics Fall 2024

Course Code	Course Title	ECTS Credits
MATH-101-03	Discrete Mathematics	6
Prerequisites	Department	Semester
MATH-180	Computer Science	Fall 2024
Type of Course	Field	Language of Instruction
Required	Computer Science/Mathematics	English
Level of Course	Lecturer	Year of Study
Undergraduate	Dr Andreas Michaelides	1 st
Mode of Delivery	Work Placement	Corequisites
Conventional		

Course Venue	Course Day and Time	Telephone
M203	Monday 10:30 – 12:00	
B18	Wednesday 10:30 – 12:00	
Office	Email	Student Consultation Hours
	Michaelides.an@unic.ac.cy	By appointment

Course Objectives:

The course covers the fundamentals of Discrete Mathematics with emphasis on applications in computing

Learning Outcomes:

After completion of the course students are expected to:

- Apply formal methods of symbolic propositional and predicate logic.
- Outline the basic structure of proof techniques, with emphasis on mathematical induction.
- Be familiar with set theory and perform the operations associated with sets.
- Identify, describe and determine the properties of relations
- Identify functions and determine their properties
- Be familiar with the concepts of graphs-digraphs-trees, model problems in computing, relate these to programming algorithms.
- Compute combinations and permutations of a set

Course Content:

1. Introduction to logic, quantifiers and conditional propositions, methods of proof, truth tables
2. Use of the first and second principle of mathematical induction in proving problems involving the set of natural numbers.
3. Set theory, set operations and algebra, switching circuits.
4. Relations and their properties, closure of relations, directed graphs, equivalence relations, partial order relations, Hasse diagrams.
5. Functions, domain, codomain and range, the properties of one-to-one and onto, composite and inverse functions, the pigeonhole principle.
6. Graphs, concepts and definitions, the travelling salesman problem
7. Trees, spanning trees, minimum spanning tree algorithm
8. Directed graphs, shortest paths, Dijkstra's algorithm for shortest path

Learning Activities and Teaching Methods:

Lectures, Exams, Homework, In-class Exercises.

Assessment:

Assessment Type	Weight (Percentage)
Final Examination	50%
Midterm Examination	30%
Graded Assignment	15%
Attendance	5%

Attendance:

You should attend all lectures. Late arrival to class will not be tolerated. Being late in class may be recorded as absence. If a student misses more than **three** classes with no excuse, he/she will be administratively withdrawn from the class. You are responsible for all material presented and all announcements made on days you are absent.

Course Requirements:

- Plagiarism in all assessment work is considered a serious offense which leads to severe punishment.
- If a student is **justifiably** absent from a Test or the Final Exam, according to the University's internal regulations, a make-up Midterm Examination may be given if **the appropriate application is filled in and the nominal fee is paid.**
- Students are expected to be cordial, courteous and respectful of faculty members and fellow students.

Grading Scale:

Letter Grade	Meaning	Numerical Grades 0-100	Grade Points 0-4
A	Excellent	93-100	4.0
A-		90-92	3.7
B+	Very Good	87-89	3.3
B		83-86	3.0
B-		80-82	2.7
C+	Good	77-79	2.3
C		73-76	2.0
C-		70-72	1.7
D+	Poor but Acceptable	67-69	1.3
D		63-66	1.0
D-		60-62	0.7
F	Failure	0-59	0.0

Required Textbooks / Readings:

Title	Author(s)	Publisher	Year	ISBN
Applied Discrete Structures	<u>Doerr</u> and Levasseur	AMS	2013	978-1-105-55929-7

Recommended Textbooks / Readings:

Title	Author(s)	Publisher	Year	ISBN
Discrete Mathematics for Computing	Haggart R	Pearson	2002	0-201-73047-2

Weekly Schedule:

Week	Topic
1	Introduction to Set Theory
2	Operations on Sets Cardinality
3	Introduction to relations
4	Properties of Relations Equivalence Relations
5	Table of Predecessors Hasse Diagram Minimal and Maximal Elements Networks
6	Midterm Examination
7	Introduction to Functions Domain and Codomain Injective and Surjective Functions
8	The inverse of a function The Pigeonhole Principle
9	Truth Tables
10	Direct Proof Proof by Induction
11	Euclid's Algorithm
12	Revision
13	Study Week
14 -15	Final Exam